Item 4: full Fourier spectrum of decompositions on the Kummer tower

2026-10-14

IC next-10 campaign, item 4 (`reports/Index calculus leads next steps.md`, "### 4."). Frozen protocol: PROTOCOL.md. Run 2026-10-14 (UTC 18:05–20:40) on the shared 4-core campaign host, single-threaded jobs under `nice` and `prlimit --as=700MB`; no isolation (load average 12–25 from the other nine items).

Verdict: NEGATIVE

C2's sparse-spectrum mechanism is closed on these toy cells by the report's kill criterion. In 31 cells (3 primes × 2 arities × 5 curves, plus the C4 curve at `t = 8`), no character's normalised coefficient on PKM's Kummer tower is above the permutation null's 99.9th percentile. This holds for the frozen statistic `Z` and for the type-standardised `Zt`. No cell exceeds on the C4 family `(1,…,1)·N/2^k` or on the marginal either. The top-100 mass `M100` is inside the null band in every cell. Three cells have a top-k mass statistic just outside the [0.1, 99.9] band; in two of the three the crossing is on the *flat* side, and none survives the pre-registered 2,000-permutation replay at `p < 0.05/31` (table below). The tower is also ordinary against the random-factor-base null: across the 31 cells, the smallest rank p of `Zt` among 100 random bases is 0.020. The positive control, a planted bias in one weighted character, is detected at `ρ = 1` in every (size, m) group, with `Zt` 3.6–23× over threshold, and down to `ρ = 0.1`–`0.3` where there are enough cross-pairs.

Scope: this closes the family on toy tower bases only, with `p ≤ 2^24`, `m ∈ {2, 3}` and `N = 2^7, 2^8, 2^11`. The statistic is the target-conditioned second moment of each character (does `c·e mod N` concentrate among one target's decompositions?). Each weighted linear form `Σ c_i e_i mod 2^k` is one character or a cyclic family of characters, and all of them were tested at once.

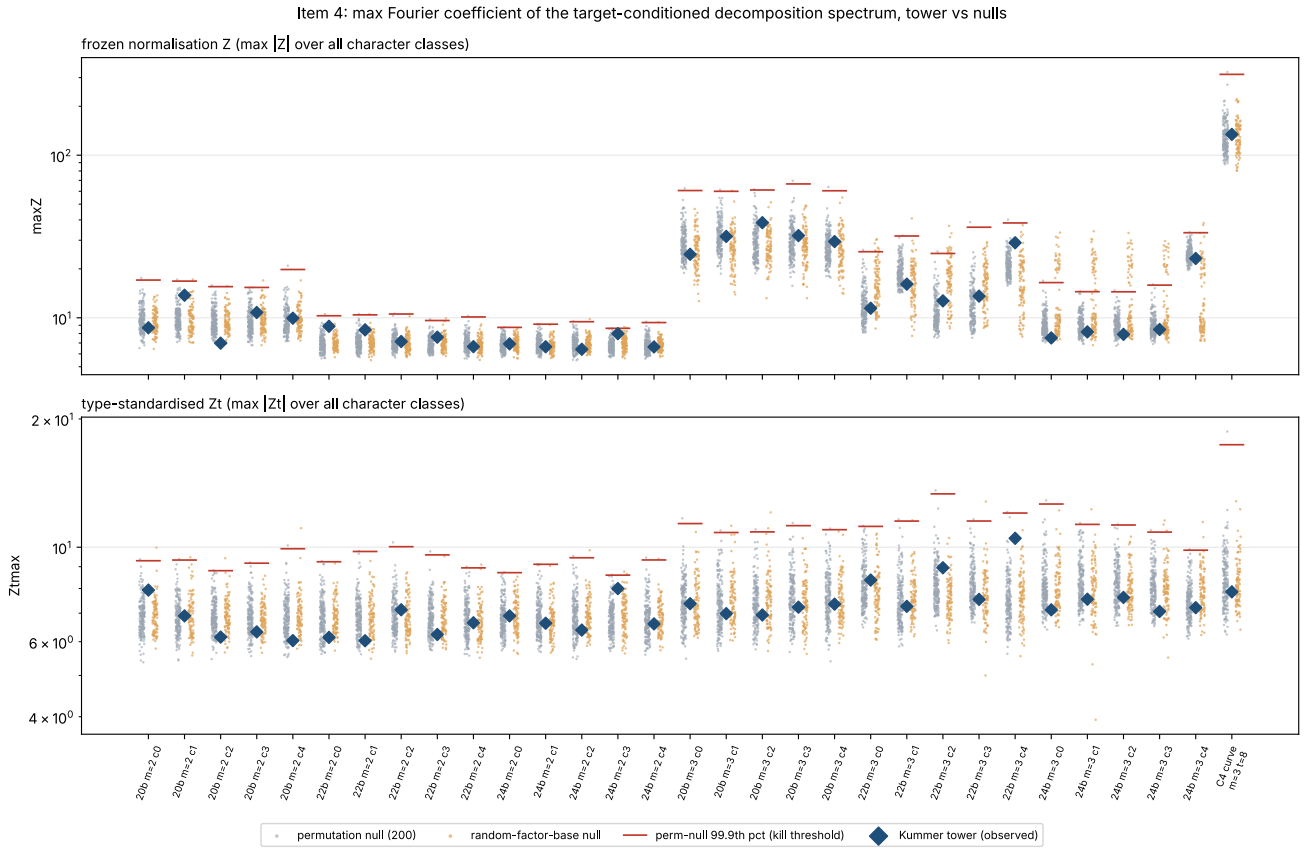
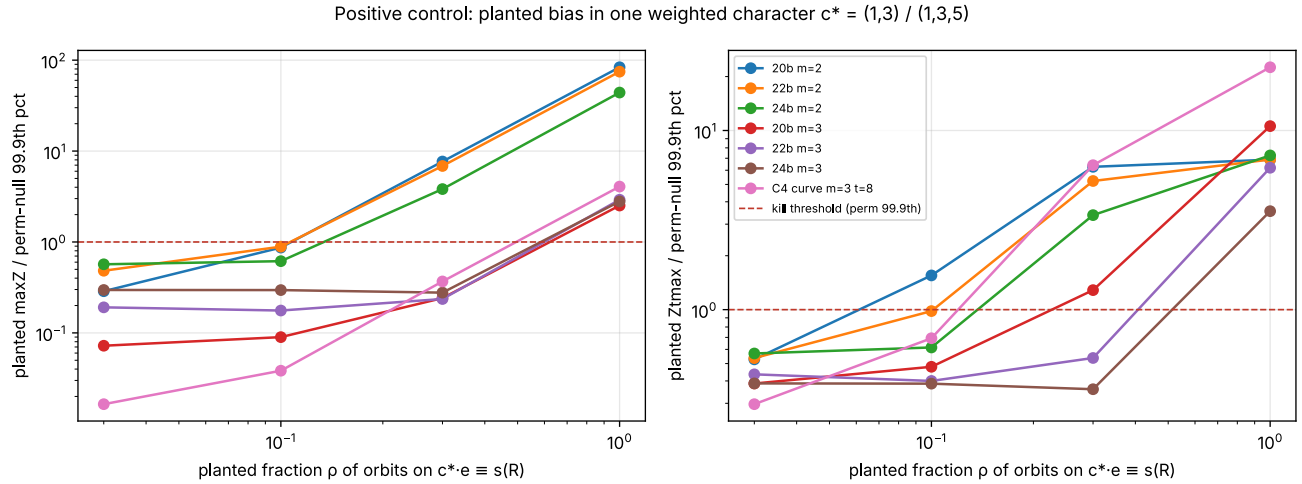


Figure 1. The largest normalised Fourier coefficient over all canonical character classes, per cell (diamond), against 200 permutation nulls (grey) and 100 random-factor-base nulls (orange); the red bar is the kill threshold (perm-null 99.9th percentile). Top: frozen `Z`; bottom: type-standardised `Zt`. Log scale. Data: `summary.csv`, `raw/`.



planted control

Figure 2. Positive control: the planted coefficient divided by the cell's kill threshold, against the planted fraction ρ (curve 0 of each group; `c* = (1,3)` for `m = 2`, `(1,3,5)` for `m = 3`). Above 1 means detected.

What was measured

Object. On the tower `V = μ_N, N = 2^t`, each base abscissa is `ζ^e`. For a target abscissa `x(R)`, the x-only decomposition indicator `f_R(e_1..e_m)` is 1 when some signs give `±P_1 ± … ± P_m = ±R`. It is `S_m`-symmetric, so it is a union of orbits, one per unordered exponent multiset. The Rust enumerator examples/ic_next10_04_kummer_fourier.rs sums **every** multiset of `m` signed base points, excluding pairs `P, −P`. Every target's decomposition set is therefore exact, and no target is sampled.

Why the unconditional spectrum is uninformative. With exhaustive enumeration, the multiset of all decomposition tuples over all targets is exactly the set of all base `m`-multisets. Its DFT depends only on which labels are on the curve, which is the "marginal" statistic. Information about the target lives in the *conditioned* spectrum:

`Â(c) = Σ_R Σ_{0≠0'⊂Z_R} f̂_0(c)·conj f̂_0'(c)`, with `f̂_0(c) = Σ_{σ∈S_m} ω^{−c·σ(e_0)}`,

which is large exactly when `c·e mod N` is predictable among one target's decompositions. It is computed for all `N^m` characters with one FFT of the `S_m`-symmetrised cross-pair difference histogram (analysis/spectrum.py).

- `Z(c) = Â(c) / (|Stab(c)|·m!·sqrt(2P))` (×√2 for real characters), where `P` is the number of cross-orbit pairs. This is the frozen normalisation.
- `Zt(c)`: `Z` standardised by the mean and sd of its zero-pattern type (1, 2 or 3 nonzero coordinates) in the same spectrum. See Deviations.
- Statistics over the canonical classes (up to `S_m` and `c ↦ −c`): `max|Z|`, `max|Zt|`, the C4 family, the top-k mass `M_k = Σ top-k Z² / Σ Z²` for `k = 1, 10, 100`, and the marginal.

Arms. These are as frozen. The tower is the observed arm. The permutation null applies a uniform random bijection of `Z/N` to the labels (200 per cell). The random-factor-base null uses `N` distinct uniform abscissas of `F_p^*`, labelled at random, with the same curve and the same enumeration (100 per cell). The positive control is a planted bias.

Results

`P` is the number of cross-orbit pairs. "≥2-orbit targets" counts targets with at least two decompositions, which are the only targets that carry information. Each threshold is the perm-null 99.9th percentile of 200 values. `p_perm` is the tower's empirical upper-tail p against the 200 permutation nulls. `p_rand` is the same against the 100 random bases.

cell	p	a	b	t	≥2-orbit targets	P	max|Z| / thr	max|Zt| / thr	p_perm(Zt)	p_rand(Zt)	C4 |Zt| / thr
20b m2 c0	1073153	695127	574028	11	310386	1011053	8.68 / 17.07	7.93 / 9.30	0.090	0.059	1.83 / 5.03
20b m2 c1	1073153	845548	665124	11	308391	993279	13.76 / 16.81	6.90 / 9.33	0.458	0.485	3.10 / 4.47
20b m2 c2	1073153	984741	114008	11	336737	1208084	6.98 / 15.55	6.15 / 8.81	0.920	0.921	2.40 / 5.67
20b m2 c3	1073153	77234	48628	11	307653	987941	10.78 / 15.37	6.33 / 9.17	0.801	0.802	1.83 / 4.38
20b m2 c4	1073153	453603	187312	11	302219	958023	9.94 / 19.82	6.04 / 9.92	0.910	0.960	1.59 / 5.13
22b m2 c0	4206593	1014557	3130927	11	209412	295234	8.87 / 10.29	6.14 / 9.24	0.816	0.941	1.82 / 5.39
22b m2 c1	4206593	2642515	2915891	11	188329	261459	8.43 / 10.43	6.03 / 9.77	0.905	0.931	2.36 / 4.25
22b m2 c2	4206593	1034562	431096	11	173769	236273	7.15 / 10.55	7.13 / 10.03	0.323	0.297	2.70 / 5.30

cell	p	a	b	t	≥2-orbit targets	P	max|Z| / thr	max|Zt| / thr	p_perm(Zt)	p_rand(Zt)	C4 |Zt| / thr
22b m2 c3	4206593	912022	903330	11	202063	283595	7.60 / 9.62	6.24 / 9.59	0.816	0.832	2.33 / 4.11
22b m2 c4	4206593	2011201	1443718	11	211134	297672	6.65 / 10.12	6.65 / 8.94	0.542	0.485	1.18 / 4.85
24b m2 c0	16801793	8809101	10887006	11	66963	73101	6.91 / 8.73	6.90 / 8.71	0.398	0.485	2.32 / 5.11
24b m2 c1	16801793	8783774	3585167	11	54168	58509	6.63 / 9.13	6.63 / 9.12	0.557	0.564	2.39 / 4.88
24b m2 c2	16801793	6629992	14532829	11	65197	70955	6.40 / 9.46	6.39 / 9.45	0.756	0.782	1.85 / 5.26
24b m2 c3	16801793	12952991	2284156	11	71596	78311	7.99 / 8.62	8.00 / 8.60	0.045	0.069	1.66 / 4.75
24b m2 c4	16801793	8232399	9078350	11	55669	60239	6.62 / 9.35	6.61 / 9.33	0.592	0.624	2.06 / 4.95
20b m3 c0	1073153	695127	574028	7	19227	23281	24.57 / 60.62	7.37 / 11.36	0.488	0.436	4.82 / 7.98
20b m3 c1	1073153	845548	665124	7	23664	29446	31.65 / 59.97	6.98 / 10.83	0.582	0.663	4.86 / 6.82
20b m3 c2	1073153	984741	114008	7	29156	37139	38.47 / 61.10	6.93 / 10.86	0.657	0.703	2.77 / 7.77
20b m3 c3	1073153	77234	48628	7	31872	41816	32.03 / 66.57	7.23 / 11.24	0.627	0.574	1.93 / 8.38
20b m3 c4	1073153	453603	187312	7	16309	19560	29.43 / 60.44	7.35 / 10.99	0.577	0.495	3.87 / 6.86
22b m3 c0	4206593	1014557	3130927	7	5284	5523	11.45 / 25.49	8.37 / 11.19	0.313	0.267	2.17 / 8.09
22b m3 c1	4206593	2642515	2915891	7	7544	7958	16.10 / 31.86	7.26 / 11.52	0.537	0.644	5.88 / 6.75
22b m3 c2	4206593	1034562	431096	7	3063	3154	12.71 / 24.87	8.95 / 13.34	0.199	0.069	3.95 / 8.40
22b m3 c3	4206593	912022	903330	7	6616	6958	13.59 / 36.06	7.54 / 11.52	0.706	0.574	4.61 / 8.74
22b m3 c4	4206593	2011201	1443718	7	9400	10007	28.95 / 38.31	10.49 / 12.03	0.015	0.020	3.66 / 7.03
24b m3 c0	16801793	8809101	10887006	7	2083	2111	7.52 / 16.46	7.13 / 12.63	0.871	0.842	1.25 / 8.56
24b m3 c1	16801793	8783774	3585167	7	881	891	8.19 / 14.47	7.55 / 11.31	0.726	0.624	1.76 / 8.31
24b m3 c2	16801793	6629992	14532829	7	1584	1604	7.91 / 14.45	7.62 / 11.28	0.706	0.584	1.82 / 8.39
24b m3 c3	16801793	12952991	2284156	7	3708	3762	8.48 / 15.88	7.07 / 10.85	0.861	0.851	2.11 / 9.84
24b m3 c4	16801793	8232399	9078350	7	1316	1320	23.17 / 33.36	7.21 / 9.84	0.617	0.752	2.60 / 5.01
C4 curve m3	946177	304973	530845	8	347665	1632009	134.24 / 314.10	7.86 / 17.40	0.642	0.614	3.07 / 8.38

Every cell has 200 permutation nulls and 100 random-base nulls. Exceedances: `max|Z|` 0/31, `max|Zt|` 0/31, C4 family 0/31, marginal 0/31.

Top-k mass bands. `M100` (frozen Z) is inside the null [0.1, 99.9] band in all 31 cells. Three cells have one top-k statistic outside its band. Each went to the pre-registered replay with 2,000 fresh permutation nulls, which needs `p < 0.05/31 = 0.0016` to count:

cell	crossing in main run (200 nulls)	replay, 2,000 fresh perms: p (side)	survives `p < 0.0016`?
20b m2 c2	`M1`, `M10` **below** band (flatter than null)	`M1` 0.0095 (lower), `M10` 0.008 (lower)	no
22b m3 c1	`ZtM100` **below** band (flatter)	`ZtM100` 0.0055 (lower)	no
22b m3 c4	`M10` above band	`M10` 0.0025, `M100` 0.004, max |Zt| 0.005, max |Z| 0.038 (upper)	no

The `22b m3 c4` cell (`p = 4206593, a = 2011201, b = 1443718`, 9,400 informative targets) is the one tower cell that sits in the upper tail on several statistics. Its random-base rank is `p = 0.020`, and its top classes are `(55,64,64)`, `(55,64,73)` and `(0,9,119)`, with negative `Zt`, i.e. *anti*-concentration. It does not reach the Bonferroni level, and the other four 22-bit `m = 3` curves are ordinary (`p_perm(Zt)` 0.20–0.71). The enumeration is exhaustive and deterministic, so this cell cannot be “re-sampled”. A check would need more curves at this size.

Under the null, about 1 % of exchangeable draws fall outside an empirical [0.1, 99.9] band of 200 values. With 31 cells × 6 mass statistics, about 1.9 crossings are expected by chance; 3 were observed.

Positive control. The planted bias is on curve 0 of each group. Detection means the planted maximum is above the cell’s kill threshold:

group	ρ = 1	0.3	0.1	0.03	smallest ρ detected (Zt)
20b m2	Z 1425 / Zt 63.9 ✔	130.9 / 58.3 ✔	14.8 ✘ / 14.4 ✔	4.9 / 4.9 ✘	0.1
22b m2	769 / 63.7 ✔	70.6 / 48.3 ✔	9.1 / 9.1 ✘ (thr 9.24)	✘	0.3
24b m2	384 / 63.3 ✔	33.3 / 29.4 ✔	5.4 ✘	✘	0.3
20b m3	152.6 / 120.2 ✔	14.6 ✘ / 14.6 ✔	✘	✘	0.3
22b m3	74.3 / 69.3 ✔	6.0 ✘	✘	✘	1.0
24b m3	45.9 / 44.8 ✔	4.6 ✘	✘	✘	1.0
C4 curve m3 t8	1278 / 392 ✔	115.6 ✘ (thr 314) / 111.4 ✔	12.1 / 12.1 ✘	✘	0.3

In every row at `ρ = 1`, the arg-max class lies in the cyclic subgroup that `c*` generates. Sensitivity scales with `sqrt(P)`. A weighted character carrying ≥ 30 % of a target’s decompositions would have been seen in every `m = 2` cell and at 20 bits for `m = 3`. At 22–24 bits with `m = 3` there are only 900–10,000 cross pairs, and only full structure (`ρ = 1`) is detectable there.

Deviations from the protocol (all disclosed)

1. **Added statistic `Zt`.** I added it after a pilot of 5–8 permutation nulls on two cells (`20b m2 c0`, `20b m3 c0`), which was run before the main data. The pilot showed that the frozen `Z` carries a label-invariant positive offset on characters with zero coordinates. When two decompositions of one target share a summand, the aligned term is 1 for every such `c`. The offset makes `max|Z|` a maximum over a few hundred shifted values, which lowers power. `Zt` standardises within each zero-pattern type. **Both statistics are reported, and the kill criterion is applied to both.** The pilot JSONs were deleted, and the main run re-ran those cells with fresh nulls.
2. **Anti-diagonal arg-max (diagnostic, post hoc).** Almost every `m = 2` cell has its `Zt` arg-max at `c = (c, −c)`. The same pattern appears in the permutation and random-base nulls (9–10 of the top 10 classes in all of them; raw/diag_antidiagonal_22_m2_c0.txt). For these characters the transposition maps `c` to `−c`, so $\hat{f}_0(c)$ is real and the cross term has twice the circular variance, which the frozen normalisation does not include. The nulls calibrate this, so the decision is unaffected, but some power is lost. A corrected normaliser is the obvious refinement for a rerun.
3. **Job chunking.** Load on the shared host reached about 25 on 4 cores, and a 200-null job would have run past the 45-minute cap. The nulls were therefore run as chunks of 50 permutations and 25 random bases, with disjoint seeds (`seed + 7919·k` for permutations, base seeds `seed·1000 + 100·k + j`). They were merged by `analysis/report.py`. The first three `20b m2` cells ran as single jobs before the change. The totals per cell are as frozen: 200 and 100.
4. **Language.** The enumeration and all curve arithmetic are Rust. The FFT, the histograms and the nulls are numpy/scipy, which the campaign `RULES.md` tool list allows. `AGENTS.md` ("Implementation language: no Python") asks for Rust in research result generation. A Rust port of `spectrum.py`, with `rustfft` (not currently a dependency), would bring this item into line.
5. **Sizes.** The report suggested `prime-kummer:t = 2,3,4`. I did not use it (frozen in the protocol): with `N ≤ 16` no target has two decompositions at `p ≥ 2^20`, so the conditioned spectrum would be empty.

Reproduction

```
cargo build --release --example ic_next10_04_kummer_fourier
export IC4_SCRATCH=/tmp/ic4 OMP_NUM_THREADS=1
S=research/ic_next10_20261014/04-kummer-fourier/analysis
python3 -I $S/spectrum.py verify          # 30/30 targets: every orbit reaches the same x(R)
# one cell (chunks k = 0..3; planted on curve 0, chunk 0)
python3 -I $S/spectrum.py cell --size 20 --m 2 --curve 0 --nperm 50 --nrand 0 --seedoff 0 --tag _perm --planted
python3 -I $S/spectrum.py cell --size 20 --m 2 --curve 0 --nperm 50 --nrand 0 --seedoff 1 --tag _perm1
python3 -I $S/spectrum.py cell --size 20 --m 2 --curve 0 --nperm 0 --nrand 25 --seedoff 0 --tag _rand
#   ... sizes 20 22 24, m 2 3, curves 0..4; continuity cell: `c4cell` with the same flags
python3 -I $S/spectrum.py cell --size 22 --m 3 --curve 4 --nperm 500 --nrand 0 --seedoff 10 --tag _replay10   # replays: seedoff 10..13
python3 -I $S/report.py   # summary.csv, summary.json, figures/
python3 -I $S/replay.py   # replay_summary.json
```

Enumerator alone: `target/release/examples/ic_next10_04_kummer_fourier --p 946177 --a 304973 --b 530845 --t 8 --m 3 --base kummer --out x.bin` prints `{"signed_points":246,"sums":2481238,"orbits":1240497,"targets":438341,"targets_ge2":347665,"cross_pairs":1632009}`.

Files

- `PROTOCOL.md`: frozen protocol.
- `analysis/spectrum.py`: spectrum, nulls and controls.
- `analysis/report.py`: merge, decision and figures.
- `analysis/replay.py`: replay p-values.
- `examples/ic_next10_04_kummer_fourier.rs`: exhaustive enumeration (Rust).
- `raw/*.json`: per-cell observed statistics, every null value and planted rows (`_perm*`, `_rand*`, `_replay*`).
- `summary.csv` and `summary.json`: per-cell decision table.
- `replay_summary.json`: replay results.
- `figures/maxcoef_by_cell.{svg,png}` and `figures/planted_control.{svg,png}`.
- `README.pdf`: this report.

What this does and does not show

It shows that on exhaustively enumerated toy Kummer-tower decompositions, no character of the exponent group `(Z/N)^m` predicts the exponents of a target's decompositions better than labels with no multiplicative meaning do. This covers the plain sum (C4), every weighted form `Σ c_i e_i mod 2^k`, and single-coordinate marginals. The second-moment spectrum is flat to the nulls' resolution, so the sparse-spectrum route of C2 has nothing to latch onto at these sizes. This also confirms the C4 falsification on the continuity cell: the C4 family `|Zt|` is 3.07 against a threshold of 8.38.

It does not show anything at `m ≥ 4`, at cryptographic `p`, or for non-character (nonlinear) structure in the exponents. It also does not address the requirements document's degree instrument $\delta_Z(1/\tilde{S})$, which is a different C2 test.